A NETWORK THEORY ANALYSIS OF JEFFREY EPSTEIN'S NETWORK

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Abstract. There exist networks of connected, powerful people that are hidden from mainstream media's line of sight, one of these being Jeffrey Epstein's network. After the release of his email data by the House Oversight Committee, we were enabled to use multiple different Network Theory algorithms and measures, to uncover the hidden, important members of this network. The network was constructed with people as nodes and weighted, directed edges as the number of emails sent and received. By running various centrality and similarity measures, we were able to uncover how important certain individuals were, what roles they played, who played similar roles to who in the overall network dynamic, and what was the overall network structure like. We found that this network was quite connected with many individuals being directly connected to each other rather than through Epstein. Then, other than Epstein himself, people such Lesley Groff, Darren Indyke, Richard Kahn, and Boris Nikolic, were people not consistently in the headlines but they made a key impacts in this network. So while the media may have portrayed Epstein as the sole dominant figure, there were certainly others behind the scenes having a significant influence.

Introduction

There has always been rumors of small networks of powerful people that make decisions which prove to be pivotal all across the world. Never has such the opportunity to peer into one of these networks existed until the slow release of email data possessed by Jeffrey Epstein. Questions were raised on how the network was arranged, who the key players were, and how communication between everyone involved behaved. With the available tools from network theory, we were able to answer these questions through the eyes of an email inbox.

Since the release of the Epstein files, while there have been many breaking news articles and massive headlines, this project is not based on any previous academic studies analyzing Epstein's network. Therefore, the results of this paper are truly our own but purely based on the network we constructed. Due to the limitations of creating a network with a raw email inbox, which will be discussed later in this paper, there are people who are left out of this network and therefore do not appear in our results, even if they have appeared frequently in news articles and in the media as an associate of Epstein.

For this project, using the tools we learned in network theory, we rigorously analyzed Jeffry Epstein's email archive to construct his infamous network. Nodes represented people within the network and directed, weighted edges represented the number of emails sent by one person to another. Epstein, of course, made all the headlines, but it was not just a one man show. By using all the standard centrality measures, such as degree, eigenvector, katz, pagerank, closeness, and betweenness centrality, we were able to uncover who, other than Epstein himself, were the core and important people to this functioning network. In addition we ran three similarity meausures, cosine similarity, the jaccard coefficient, and hamming distance, to uncover who player similar roles within the network. Essentially, the main goal of this project was to truly uncover and reveal the functioning of this pervasive network.

Network Analysis Background

The following formulas on network centrality and node similarity are tools from network theory we utilized to help us draw conclusions about this network. In network theory, the centrality of a node tells one how important a node is in the network. The one thing to note is that different centrality measures use different definitions of important. For degree centrality, the calculation is simple. The centrality x_i of node i is equal to its degree, thus giving the following equation

$$x_i = k_i \tag{1}$$

where k_i is the degree of node i; the number of edges connected to it. Degree centrality essentially measures the importance of a node by the number of nodes it is connected to.

On the other hand, eigenvector centrality does not care if a node is connected to many other nodes. It instead cares if a node has important neighbors. The formula for eigenvector centrality is given below:

$$x_i = \frac{1}{r} \sum_{j=1}^{n} A_{ij} x_j \tag{2}$$

where r is a constant and it is the largest eigenvalue of the adjacency matrix A and x_j is the eigenvector centrality of node j.

Katz centrality is very similar to eigenvector centrality as the formula is virtually the same except Katz centrality gives all nodes some amount of centrality "for free." This is to avoid nodes having a zero eigenvalue and instead give them some centrality score. The formula is as follows:

$$x_i = \alpha \sum_{j=1}^{n} A_{ij} x_j + \beta \tag{3}$$

The main difference between equation (2)and (3) is the added β constant at the end. This represents the centrality that all nodes get "for free." It is important to note the α constant in front of the summation replaced $\frac{1}{\lambda_1}$ where the α constant is constrained to be striclty between 0 and $\frac{1}{\lambda_1}$. This is because in the Katz centrality vector equation shown below

$$\boldsymbol{x} = \alpha A \boldsymbol{x} + \beta \mathbf{1} \tag{4}$$

to solve for $\boldsymbol{x}$, the matrix $(I - \alpha A)$ must be invertible which is only the case when α follows that constraint.

Within a directed network, PageRank centrality is quite similar to Katz centrality but it addresses the issue that if a node within a network has a very high Katz centrality but points to many other nodes, all those nodes it points to get a high centrality score just by association. Practically speaking, this may not be a realistic representation of a nodes importance if it is classified as important just because it is one of thousands to point to an important node. As a solution, PageRank centrality dilutes the importance of a neighbor if it points to many others. The equation is as follows:

$$x_i = \alpha \sum_{j=1}^{n} A_{ij} \frac{x_j}{k_j^{out}} + \beta \tag{5}$$

Notice the $\frac{x_j}{k_j^{out}}$ within the equation represents how having an important neighbor gets diluted if k_j^{out} is really high, (i.e. node j has a really high out-degree).

The next centrality measures are closeness and betweenness centrality. Unlike the previous four centrality measures that measure a nodes importance based on information about the it's neighbors, these are path based. Closeness centrality classifies nodes as important if they have a small distance to many other nodes. It is defined as

$$x_i = \frac{1}{\ell_i} \tag{6}$$

where ℓ_i is the mean shortest path length between node i and any other node. The equation for ℓ_i is as follows:

$$\ell_i = \frac{1}{n}\sum_{j=1}^{n} d_{ij} \tag{7}$$

where d_ij is the length of the shortest path between nodes i and j.

Then, betweenness centrality says that a node is important if it lies on the shortest path between many other pairs of nodes. It is defined as

$$x_i = \sum_{s=1}^{n}\sum_{t=1}^{n} n_{st}^{i} \tag{8}$$

where

$$n_{st}^{i} = \begin{cases} 1 & \text{if node } i \text{ lies on shortest path between nodes s and t} \\ 0 & \text{otherwise} \end{cases}$$

In addition to the centrality measures above, we also computed three similarity measures across all nodes. In network theory, similarity identifies pairs of nodes that may share a lot of neighbors or nodes that may play a similar role in the network, despite not even being connected. The first measure is cosine similarity, which takes it a step further than automatically saying two nodes are very similar if they share the same neighbors. It is defined as

$$\cos(i,j) = \frac{n_{ij}}{\sqrt{k_i k_j}} \tag{9}$$

where n_{ij} represents the number of common neighbors between nodes i and j. By dividing by the square root of $k_i k_j$, cosine similarity takes into account the degree of each node to normalize the value of the number of common neighbors. For example, having 5 common neighbors may not seem like a lot but if both nodes only have a degree of 5, then they'll have a cosine similarity of 1. Thus, this measure can be used to identify two nodes that occupy structurally analogous positions within the network, regardless of whether they are directly connected.

The second measure is the jaccard coefficient which measures the fraction of common neighbors two nodes share.

$$J_{ij} = \frac{n_{ij}}{k_i + k_j - n_{ij}} \tag{10}$$

A jaccard coefficient of 1 implies that two nodes share all neighbors while a jaccard coefficient of 0 implies that two nodes share no common neighbors.

The third measure, hamming distance, measures the number of neighbors not shared by two nodes.

$$h_{ij} = \sum_{k=1}^{n}(A_{ik} - A_{jk})^2 \tag{11}$$

When node k is a neighbor of both nodes i and j, $(A_{ik} - A_{jk})^2 = (1-1)^2 = 0$. When it is only a neighbor of one but not the other, $(A_{ik} - A_{jk})^2 = (0-1)^2 = 1$ or $(A_{ik} - A_{jk})^2 = (1-0)^2 = 1$. And when k is not a neighbor of either, $(A_{ik} - A_{jk})^2 = (0-0)^2 = 0$. Thus, the hamming distance only increases when nodes i and j do not share common neighbors.

Model and Methodology

The network was constructed from Epstein's email archive, which was released through the House Oversight Committee, and it was made possible to access through the Jmail Data API. The data downloaded was approximately 1.8 million raw email records. The raw data had fields for `to`, `cc`, and `bcc` recipient lists, a timestamp, and a promotional-email flag. After all promotional emails were filtered out, and retaining only records involving at least one identified inner-circle member, approximately 904,000 rows were used in the final analysis. The primary data engineering challenge was that a single individual frequently appeared under dozens of different email addresses and name spellings. Epstein alone had more than sixteen distinct sender identities and aliases across the archive. To address this duplication problem, we constructed two hand-curated lookup tables:

`NAME_MAP`

and,

`EMAIL_MAP`

Where the name map, is used to map raw sender name strings to canonical full names, and the email map is used to map full names to sender addresses. We then defined 53 confirmed inner-circle individuals, who then became the nodes of our network. Any sender or recipient not mapping to a whitelist entry was removed from the dataset. It needs to be made clear that the selection of the 53 members of this network was done with subjective discretion and judgment, where only those who were selected were those that could be seen in photos, headlines, or text messages, with Jeffery Epstein. Any email aliases that couldn't be confirmed with certainty were discarded.

From the cleaned data we constructed a directed, weighted graph,

$$G = (V, E, w)$$

implemented as a `NetworkX DiGraph`. Each of the 53 confirmed individuals were a node in the set V. A directed edge represented as

$$(i \to j) \in E$$

existed only if person i sent at least one email in which person j appeared as a recipient. And then the edge weight was the total count of such emails across the `to`, `cc`, and `bcc` fields represented as

$$w(i, j) \in \mathbb{Z}^+$$

Self-loops, as in someone sent an email to themselves, were discarded. Nodes from the whitelist with no edges in the filtered data (isolates) were also removed, leaving 53 nodes and 555 directed, weighted edges.

It is important to note however that certain methodologies in our construction need flagging. First, the edge weight treats `to`, `cc`, and `bcc` recipients identically: a node that was blind-copied on 200 emails accumulates the same weight as one who received 200 direct messages. Second, weights are raw email counts with no normalization for time, thread structure, or message length, so a 500-message reply chain is treated identically to 500 standalone communications.

All centrality measures were computed using `NetworkX`. Clustering coefficient and undirected degree centrality are defined only on undirected graphs, so for those two measures we used the undirected projection G^{und}, obtained by symmetrizing G and collapsing reciprocal directed edges into a single undirected edge.

For Katz centrality, an importnat thing to note is the parameter α was set to the following:

$$\alpha = \frac{0.9}{\lambda_1}$$

The factor of 0.9 places α near but strictly inside the convergence boundary. The baseline score β was left at the `NetworkX` default of 1.0.

For the small-world analysis, we compared the actual network against 100 Erdős–Rényi random graphs $G(n, p)$ with $n = 53$. Since clustering coefficient and average path length are defined only on undirected graphs, both measures were computed on the undirected projection G_{und}. The ER baseline was therefore matched to the undirected edge density rather than the directed density. These differ because symmetrizing the graph collapses reciprocal directed edges into single undirected edges while halving the maximum possible edge count. The formulas below are the density calculations for the directed and undirected networks.

$$\rho_{\text{directed}} = \frac{|E|}{n(n-1)} = \frac{555}{53 \times 52} = \frac{555}{2756} \approx 0.2014 \tag{12}$$

$$\rho_{\text{undirected}} = \frac{|E_{\text{und}}|}{n(n-1)/2} = \frac{314}{53 \times 52/2} = \frac{314}{1378} \approx 0.2279 \tag{13}$$

Only connected ER realizations were retained, as average path length is undefined for disconnected graphs. The small-world coefficient σ is defined as

$$\sigma = \frac{C/\langle C_{\text{rand}}\rangle}{L/\langle L_{\text{rand}}\rangle} \tag{14}$$

where C and L are the clustering coefficient and average path length of the actual network, and $\langle C_{\text{rand}}\rangle$ and $\langle L_{\text{rand}}\rangle$ are the means over the 100 random graph replicates. A network is considered small-world when $\sigma > 1$. This network achieves $\sigma = 3.154$.

Results

The following results are based on the directed, weighted graph constructed from Epstein's email archive, filtered to the 53 confirmed inner-circle individuals. Table 1 summarizes the structural properties of the network.

Table 1. Summary graph properties for the full network.

Property	**Value**
Graph type	Directed, weighted
Nodes ($\|V\|$)	53
Directed edges ($\|E\|$)	555
Epstein weighted in-degree	143,716
Epstein weighted out-degree	143,818
Avg. clustering coefficient $\bar{C}$	0.6897
Avg. path length L	1.7721
Small-world σ	3.154
Denisty	0.2014

The network is moderately dense at $\rho = 0.2014$ meaning roughly one in five possible directed edges between inner-circle members exists. Epstein's symmetric weighted degree immediately establishes him as the dominant hub. The average path length of $L = 1.77$ is extremely short, suggesting that the overall network was very well-connected where essentially any two members could reach each other within a couple of emails.

In looking at Table 2 which records the top 10 people in terms of node in- and out-degree along with their real-life role, Epstein's raw email volume dwarfs all other nodes, sending and receiving roughly 2.7 times as many emails as the second-highest node, Lesley Groff, in each direction. The asymmetry in Groff's degree, with 52,610 emails sent vs. her 26,768 received suggests she was consistently initiating communication rather than responding. Ann Rodriquez appears exclusively on the out-degree side of the top 10, indicating she was an active sender yet received relatively little direct correspondence back. This is consistent with her role as a travel planner in sending emails about the logistics of upcoming events. Darren Indyke appears exclusively on the in-degree side, which is also consistent with his role as the primary manager of Epstein's legal affairs.

Then, Table 3 summarizes the top 10 people in betweenness centrality. The most striking result, probably in the entire analysis is that Indyke, being Epstein's personal lawyer and estate executor, achieves a betweenness centrality of 0.38, nearly double that of Ghislaine Maxwell and higher than Epstein himself

Table 2. Node Degree Rankings

Rank	**Node**	**In-Degree**	**Out-Degree**	**Role**
1	Jeffrey Epstein	143,716	143,818	—
2	Lesley Groff	26,768	52,610	Executive assistant
3	Richard Kahn	22,922	22,385	Chief of staff
4	Boris Nikolic	10,702	8,124	Harvard-affiliated scientist
5	Karyna Shuliak	10,194	8,917	Personal staff
6	Darren Indyke	7,219	—	Personal lawyer, estate executor
7	Bella Klein	7,168	6,740	Personal staff
8	Larry Visoski	6,849	7,585	—
9	Ann Rodriquez	—	8,526	Travel manager
10	Tazia Smith	—	5,086	Portfolio management

Table 3. Betweenness Centrality Rankings

Rank	**Node**	**Betweenness**
1	Darren Indyke	0.3798
2	Ghislaine Maxwell	0.1958
3	Kathy Ruemmler	0.1928
4	Boris Nikolic	0.1558
5	Ariane de Rothschild	0.1510
6	Richard Kahn	0.1422
7	Lesley Groff	0.1357
8	Jes Staley	0.1303
9	Karyna Shuliak	0.1231
10	Bella Klein	0.1047

who's not even in the top 10. This indicates that Indyke was a sort of bridge, sending and receiving emails to and from many otherwise disconnected people in the network. Kathy Ruemmler's betweenness of 0.193, placing her third overall, is anomalous relative to her low raw degree. She does not appear in the top 10 for raw email volume in Table 2, yet she, similarly to Indyke, bridges otherwise disconnected parts of the network. This makes sense given that she was a secondary legal advisor for Epstein's circle, a smaller role compared to Indyke.

Table 4. Eigenvector, Katz, and PageRank Centrality

Node	**Eigenvector**	**Katz**	**PageRank**
Jeffrey Epstein	0.7264	0.6840	0.3752
Lesley Groff	0.3813	0.3536	0.0748
Richard Kahn	0.3329	0.3120	0.0586
Boris Nikolic	0.2243	0.2160	0.0264
Karyna Shuliak	0.1472	0.1514	0.0232
Larry Visoski	0.1077	0.1172	0.0164
Kathy Ruemmler	0.1009	0.1114	—
Darren Indyke	0.0823	0.0964	0.0169

Table 4 summarizes the top 8 individuals in Eigenvector, Katz, and PageRank centrality. Interestingly enough, all three centrality measures agree on the same chain of command: Epstein $\rightarrow$ Groff $\rightarrow$ Kahn $\rightarrow$ Nikolic. The consistency of this ranking across three distinct centrality formulations strongly indicates this hierarchy is robust and not an artifact of any particular measure.

Ruemmler appears again, this time at rank 7 in eigenvector centrality. This once again reinforces the idea that she communicated primarily with very important individuals like Epstein. Indyke presents the sharpest contrast in the dataset, having an extremely high betweenness centrality but a comparatively low eigenvector centrality. This pattern seems to suggest that he controlled correspondence between clusters of individuals without being a primary communication partner of the top-degree nodes.

Table 5. Closeness Centrality Rankings

Rank	**Node**	**Closeness**
1	Jeffrey Esptein	1
2	Lesley Groff	0.9630
3	Richard Kahn	0.7324
4	Karyna Shuliak	0.6667
5	Darren Indyke	0.6500
6	Bella Klein	0.6265
7	Daphne Wallace	0.6047
8	Cecile de Jongh	0.5778
9	Larry Visoski	0.5778
10	Brice Gordon	0.5714

Table 5 then summarizes the top 10 people in closeness centrality. Epstein achieves perfect closeness centrality of 1, consistent with the confirmed structural property that he maintained a direct edge to all 52 other nodes. Groff's closeness of 0.963 is nearly as high, indicating she too was reachable from almost every other node in a single email. This is consistent with the very short average path length established in Table 1.

Table 6. Network Metrics: Full Network vs. Epstein Removed

Metric	**Full Network**	**Epstein Removed**
Nodes	53	52
Edges	555	451
Avg. clustering (undirected)	0.6897	0.5886
Top betweenness node	Darren Indyke (0.380)	Darren Indyke (0.368)
Top PageRank node	Jeffrey Epstein (0.375)	Lesley Groff (0.196)

Table 6 takes an interesting perspective and compares the full network versus one without Epstein. Removing Epstein from the graph eliminates 104 directed edges. This is a fairly small change and the residual network remains a single connected component which helps support the notion that this was a social circle where the individuals formed connections independent of Epstein.

The average clustering coefficient of $\bar{C} = 0.69$ is the first indication of small-world structure. When any two nodes share a common email correspondent, there is a 69% chance they directly communicate with each other. Just to note, several peripheral nodes achieve a perfect local clustering coefficient of 1.0 — including Ramsey Elkholy, Natalia Molotkova, Tazia Smith, and Eric Roth — indicating that these individuals exist entirely within their own complete sub network where every member is in direct contact to every other member. Interestingly enough, the average clustering coefficient for the network without Epstein does fall, indicating that Epstein, especially given his closeness centrality of 1, was a key connector in this network. However it only drops from around 0.69 to around 0.59, indicating that the network without him was still quite connected.

Even after Epstein's removal, Indyke's betweenness centrality remains that highest and only falls from 0.380 to 0.368. This confirms that his role is not contingent on Epstein acting as an intermediary. On the other hand, the most dramatic shift is in pagerank centrality. Groff's score jumps from 0.075 with Epstein to 0.196 without, making her the dominant node in the residual network. In the absence of Epstein she becomes the clear hub through which the rest of the network is organized which comes as no surprise given her previous ranking in eigenvector, katz, and pagerank centrality and the fact that her closeness centrality was very close to 1.

Table 7. Small-world analysis

Metric	**Actual Network**	**Erdős–Rényi (mean)**	**Ratio**
Clustering coefficient	0.6897	0.2252 ± 0.018	$3.06\times$
Average path length	1.7721	1.8252 ± 0.028	$0.97\times$
Small-world σ	3.1541	1.0 (by definition)	—

Table 7 records the small-world analysis of this network. The network achieves clustering 3.06 times higher than a comparable random graph while maintaining an average path length just around the random baseline. This is a clear indication of a small-world network, and reinforces what we have found regarding the tightly-woven social circle that the network represents.

Moving on to similarity, we constructed three 53 x 53 heat maps for the three different similarity measures, xosine similarity, the jaccard coefficient, and hamming distance. Each cell is visual representation of the similarity between two nodes where the darker the cell, the more similar they are, and the lighter the cell, the less similar they are for the respective similarity measure. These heat maps are cited in the Appendix.

The cosine similarity heatmap in Appendix B reveals a striking visual feature. Epstein and Groff each produce a distinctly lighter band running across the full heatmap but a very dark cell at their intersection. This pattern is present because both of them are uniquely powerful members of the network. Their connections are much more exhaustive than any other node, making them dissimilar to nearly everyone except each other. Interestingly, it seems that high centrality produces low cosine similarity with most of the network since a node connected to everyone is a very unique quality (i.e. Epstein and Groff). The remaining nodes, being much less globally connected, tend to have higher similarity to each other in that they are much less structurally important than big nodes like Epstein and Groff. A secondary finding is the elevated cosine similarity between Indyke and Ruemmler, which is consistent with both individuals serving as managers for Epstein's legal affairs. Their connection profiles overlap because they were likely communicating with the same subset of relevant contacts.

Table 8. Jaccard Coefficient

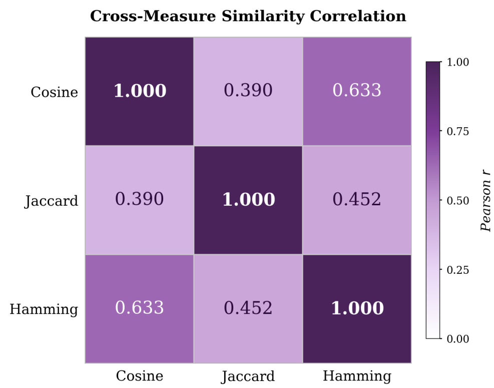
Rank	**Node i**	**Node j**	**Jaccard Coefficient**
1	Juan Pablo Molyneux	Ramsey Elkholy	1
2	Jeffry Epstein	Lesley Groff	0.9434
3	Stewart Oldfield	Vahe Stepanian	0.7692
4	Brad Wechsler	Richard Joslin	0.7500
5	Jes Staley	Nicole Junkermann	0.7500
6	Paul Morris	Vahe Stepanian	0.7143
7	Lesley Groff	Richard Kahn	0.6923
8	Jeffrey Epstein	Richard Kahn	0.6792
9	Faith Kates Peggy	Siegal	0.6667
10	Tazia Smith	Vahe Stepanian	0.6667

The jaccard coefficient heatmap in Appendix C reinforces the visuals observed with Epstein and Groff in cosine similarity, with the two once again ranking among the highest-similarity pairs, indicating that nearly every connection one of them maintained was also maintained by the other. Given Groff's role as executive assistant, it is expected that she would mirror Epstein's contact list almost entirely. An unexpected finding is that Juan Pablo Molyneux and Ramsey Elkholy achieve a jaccard coefficient of 1.0 as shown in Table 8, meaning they shared the exact same email correspondents. This is almost certainly a consequence of both individuals having limited connection sets, likely confined to Epstein, each other, and one or two additional contacts. With so few email connections, complete overlap is much more achievable and the result suggests

Table 9. Normalized Hamming Distance Scores (Between 0 and 1)

Rank	Node i	Node j	Hamming Distance
1	Juan Pablo Molyneux	Ramsey Elkholy	1
2	Petra Nemcova	Ramsey Elkholy	0.9811
3	Juan Pablo Molyneux	Petra Nemcova	0.9811
4	Jes Staley	Nicole Junkermann	0.9623
5	Faith Kates	Peggy Siegal	0.9623
6	Larry Summers	Ramsey Elkholy	0.9623
7	Juan Pablo Molyneux	Larry Summers	0.9623
8	Jeffrey Epstein	Lesley Groff	0.9434
9	Stewart Oldfield	Vahe Stepanian	0.9434
10	Brad Wechsler	Richard Joslin	0.9434

Figure 1. Pearson Correlation Matrix Across The Three Similarity Measures



these two were peripheral figures who interacted with the inner circle exclusively through a common narrow channel rather than as independent members.

The hamming distance heatmap in Appendix D once again identifies Epstein and Groff as structurally equivalent, consistent with the two previous similarity measures. The rankings in Table 9 once again involve Molyneux, Elkholy, and a third individual, Petra Nemcova, who together form the highest-hamming cluster in the network. This strongly suggests a tight clustering among these three that imply not just that they shared common email connections, but that they were all similarly isolated from the rest of the network. This could be an indication of a clique operating at the periphery of the network: a small group of individuals who are well-connected to each other and to a narrow set of shared contacts, but who have little to no independent presence in the broader network.

Then, in looking at how the three similarity measures compare, Figure 1 displays the Pearson Correlation Matrix across them. Across the three, cosine similarity and hamming distance exhibit the highest pairwise correlation at $r = 0.633$. The reason for the higher similarity between these two in particular can be derived from the heatmaps. Both have distinctly lighter portions of the grid that clearly indicate where Epstein and Groff lie in the chart whereas the jaccard coefficient heatmap does not show it nearly as well. This may be because peripheral nodes do not necessarily share common email correspondents. Some may be in direct contact with Epstein and Indyke, while others may only be in direct contact with Groff. This would lead to a low jaccard coefficient but high cosine similarity and high Hamming Distance since these peripheral nodes play similar roles in the network and do not share a lot of email correspondents. Note that this is only speculation but the logic does align with the results.

Conclusions and Discussion

In this project, we constructed a directed, weighted network using Jeffrey Epstein's email archive and analyzed the people in it using several centrality and similarity measures from network theory. By treating individuals as nodes and emails as directed edges, we were able to study how communication was structured and identify which individuals played the most important roles in the network. Instead of relying on just one definition of importance, we used multiple centrality measures—degree, eigenvector, katz, pagerank, closeness, and betweenness, as well several similarity measures–cosine, jaccard coefficient, and hamming distance, to get a more complete picture of this network.

As expected, Epstein dominates the network across nearly all measures, but the results show that the network is far from a one-person system. Individuals like Lesley Groff and Richard Kahn consistently rank highly, especially in degree and eigenvector-based measures, meaning they are both highly connected and connected to other important nodes or individuals within this context. At the same time, Darren Indyke stands out in terms of betweenness centrality, suggesting that he plays a key role as a bridge between different parts of the network. This is important because it shows that the most "important" person truly depends on how importance is being defined; some individuals are central because of how many connections they have, while others are central because they control the flow of information.

Looking at the overall structure, the network has strong small-world properties with a high clustering coefficient and short average path length. This means that most individuals are closely connected, and information can spread efficiently through the network. When Epstein is removed, the network becomes less dense and slightly less connected, but it does not completely fall apart. Instead, other nodes, especially Groff, become more central. This suggests that while Epstein is clearly the dominant node, the network still has some level of structure and does not rely entirely on a single individual.

One thing that worked well in this project was using multiple centrality measures since each one captures a different aspect of importance. This made it easier to identify not just who is most connected, but also who plays key roles in linking different parts of the network. The same goes for similarity as using all three measures gave a better picture on who truly was similar to who. However, there are also some limitations to this project. The network is based only on email data, so it likely does not capture all real-world interactions. Also, centrality does not necessarily translate directly to real-world influence, so the results should be interpreted carefully.

If we had more time, there are a few ways this project could be improved. For example, we could look at how the network changes over time instead of treating it as static. We could also apply community detection methods to identify subgroups within the network, or compare this network to other similar datasets to better understand how unique it is.

Overall, this project shows how network theory can be used to analyze complex communication systems and identify key individuals within them. Although a network may appear to be centered around one person, it is revealed that there are multiple important figures who help shape its structure and function in various ways after analyzing deeper.

References

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Acknowledgments

We would like to thank Dr. Tymochko for allowing us to go through with this project.

Appendix A. Group Contributions

Conor McDermott: Data and Model Engineering; Author : Introduction, Model and Methodology
Danilo Talisayon: Author : Introduction, Background
Joseph Koslosky: Author : Results
Brennan DeCourcey: Author : Conclusions and Discussion

Appendix B. Cosine Similarity Heat Map

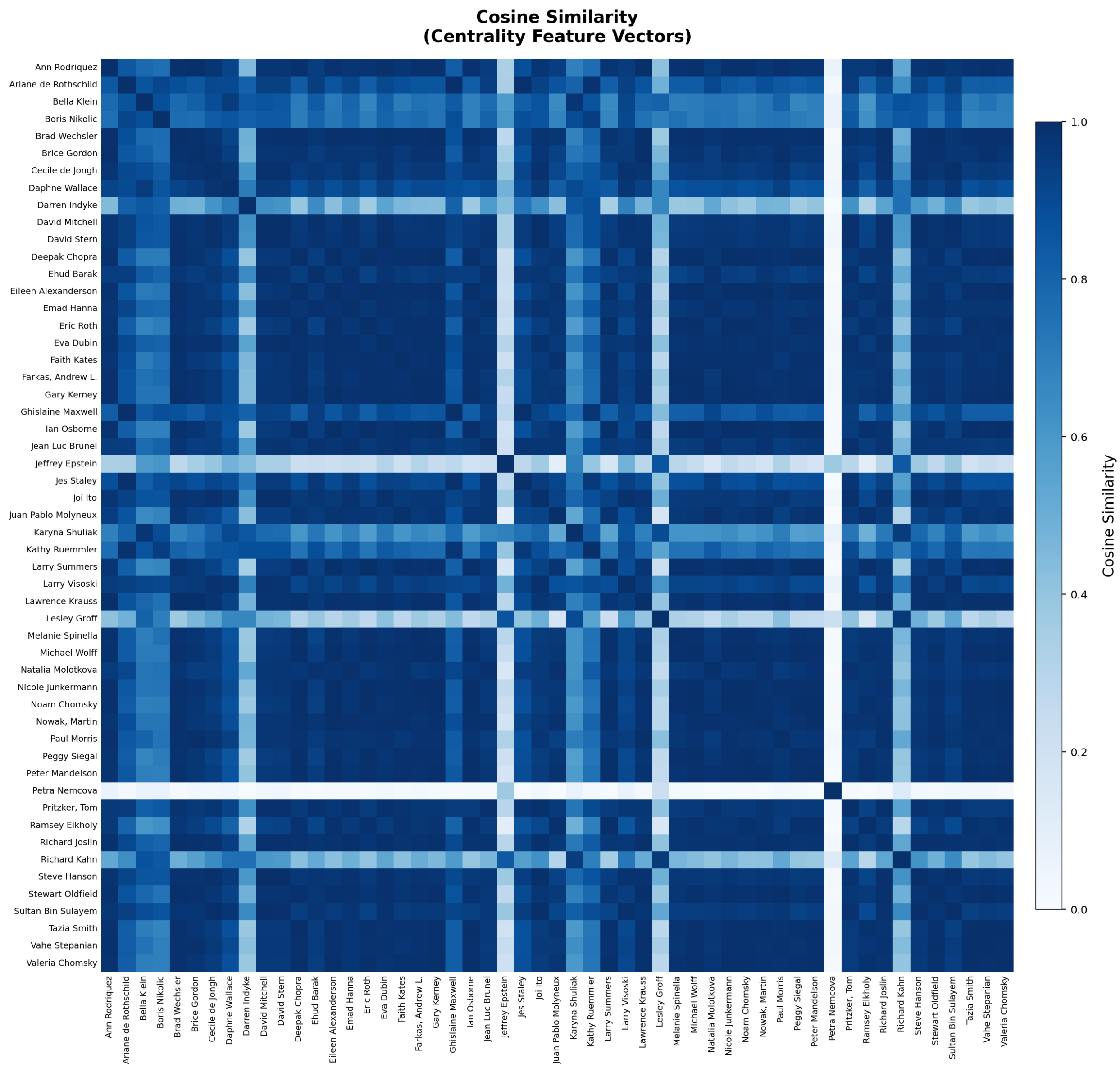

Figure 2. Cosine similarity for all 53 nodes. High values indicate nodes that share similar roles in the network regardless of direct connection. The diagonal equals 1.0 by construction.

Appendix C. Jaccard Coefficient Heat Map

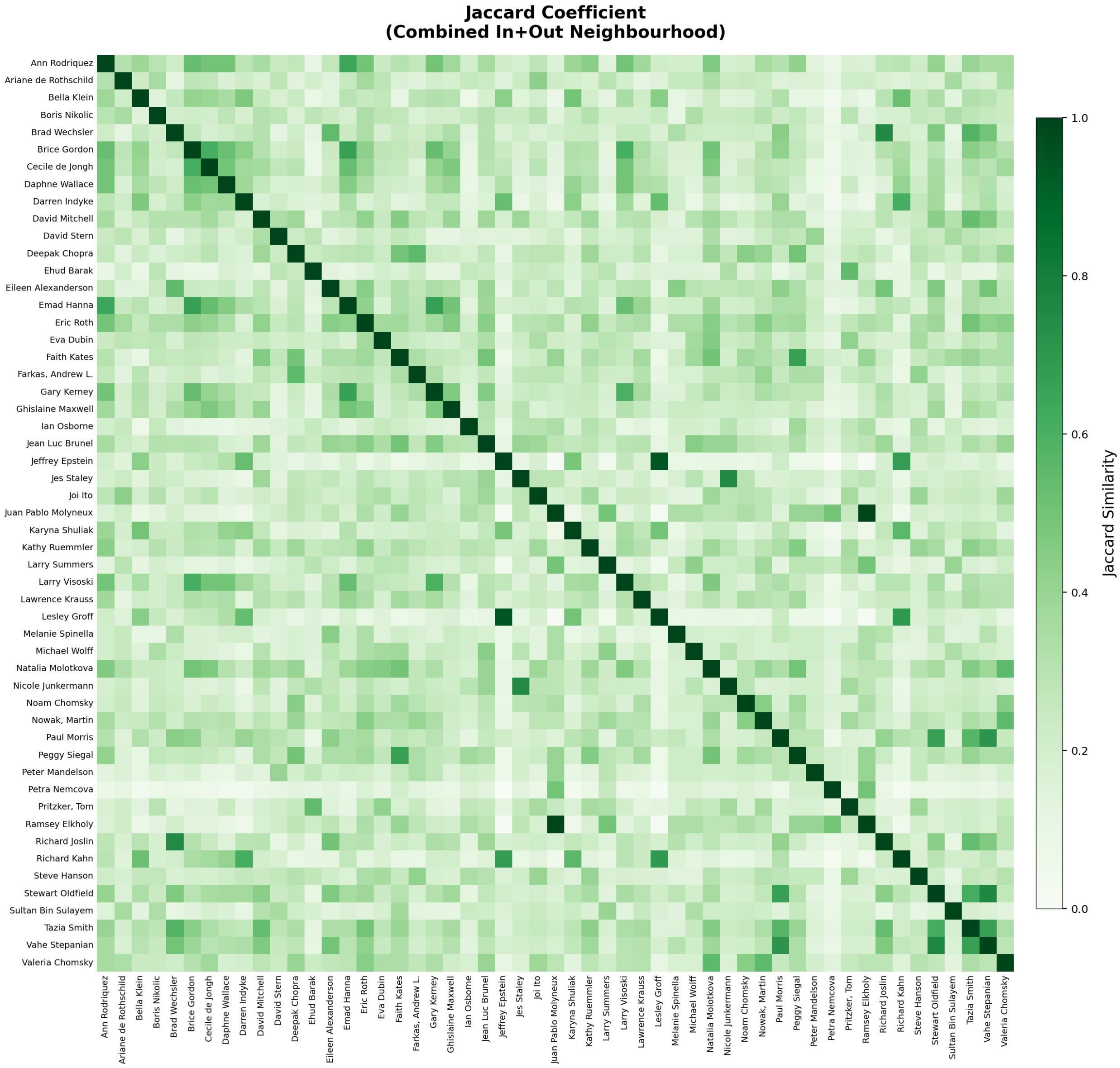

Figure 3. The Jaccard coefficient for all 53 nodes. A value of 1.0 indicates completely identical email correspondents. Note that two lesser know figures, Juan Pablo Molyneux and Ramsey Elkholy, they achieve a perfect score of 1.0. The diagonal equals 1.0 by construction.

Appendix D. Hamming Distance Heat Map

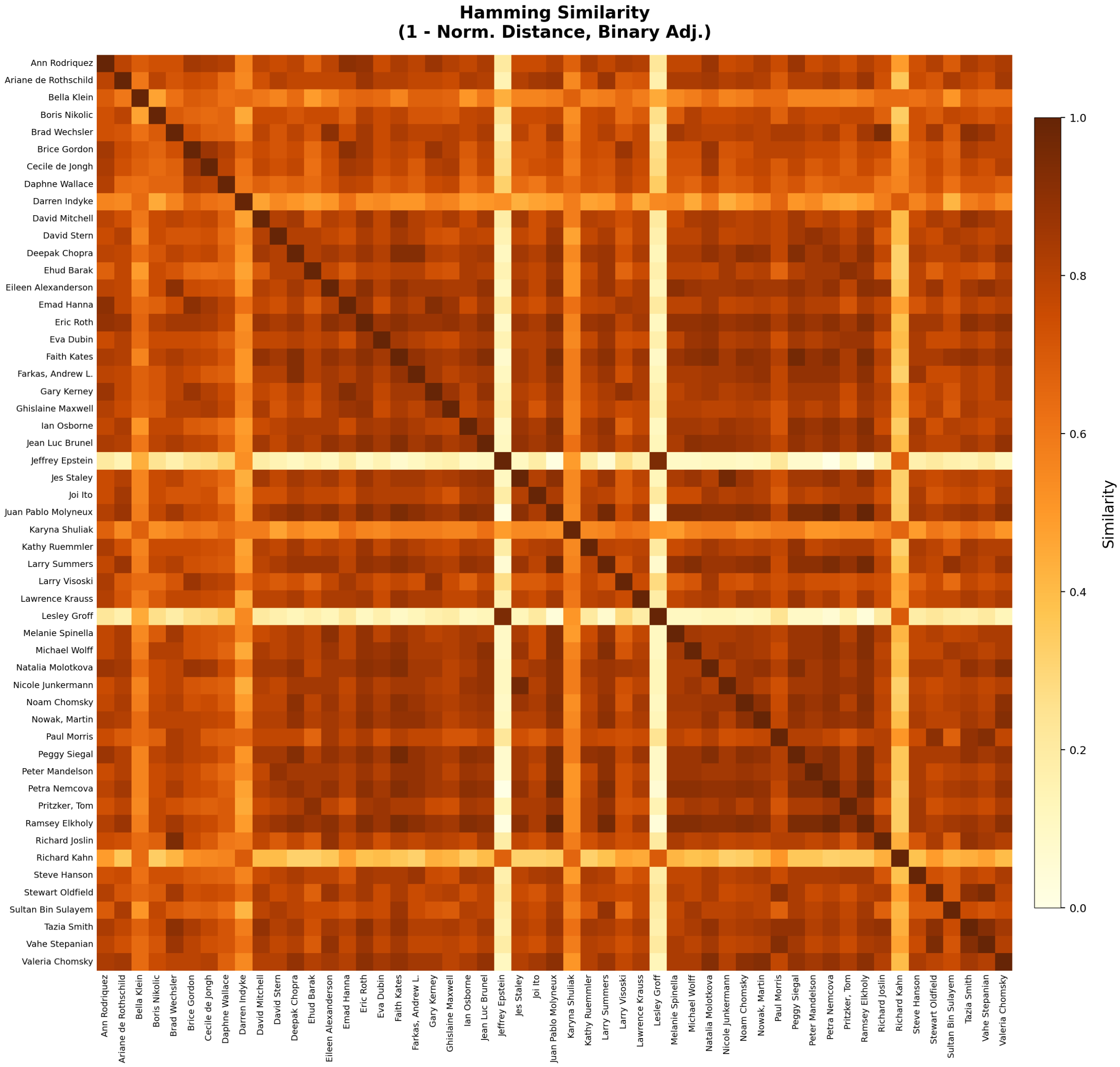

Figure 4. Hamming Distance for all 53 nodes. High values indicate nodes with nearly identical email connections across the network. The diagonal equals 1.0 by construction.